NPTEL Online Certification Courses

Indian Institute of Technology Kharagpur

Introduction to Internet of Things

Assignment-Week 12

TYPE OF QUESTION: MCQ/MSQ

Number of questions: 15 **Total marks: 15 X 1= 15**

QUESTION 1:

Qualitative analysis does not deal with the analysis of data that is categorical in nature.

a. True

b. False

Correct Answer: b. False

Detailed Solution: Qualitative analysis deals with the analysis of data that is categorical in nature. (Please refer Lecture 56@2:43)

QUESTION 2:

Fill in the blank.

Examples of dispersion measures include _____ and _____.

a. Discrete, categorical
b. Continuous, quantitative
c. Discrete, quantitative
d. Range, Variance

Correct Answer: d. Range, Variance

Detailed Solution: Examples of dispersion measures include Range and Variance (Please refer Lecture 56@13:51)

QUESTION 3:

Select the statement(s) that denote the key variables of determining statistical significance.

Statement I: Range size

Statement II: Effect size

Statement III: Sample size

a. Statement I

b. Statement I and II

c. Statements II, and III

d. None of these

Correct Answer: c. Statements II, and III

Detailed Solution: The key variables of determining statistical significance include effect size and sample size (Please refer Lecture 56@16:32)

QUESTION 4:

The different objectives of AgriSens includes –

a. More yields with less water

b. Save limited water resource

c. Automatic irrigation

d. All of these

Correct Answer: d. All of these

Detailed Solution: The different objectives of AgriSens includes More yields with less water, Save limited water resources and Automatic irrigation. (Please refer Lecture 57@5:32)

QUESTION 5:

The two most relevant sensors directly used in agriculture are _____.

a. Soil moisture and ECG sensor

b. Soil moisture and water level sensor

c. ECG sensor and water level sensor

d. All of these

Correct Answer: b. Soil moisture and water level sensor

Detailed Solution: Soil moisture and water level sensors are the necessary sensors generally used in agriculture. (Please refer Lecture 57@13:11)

QUESTION 6:

The integrated design of the remote server includes_____.

a. Data server

b. Web server

c. Multi users server

d. All of these

Correct Answer: d. All of these

Detailed Solution: The integrated design of the remote server includes repository data server, web server and multi-users server. (Please refer Lecture 57@9:35)

QUESTION 7:

Select the correct order of the component layers present in the IoT healthcare.

a. Sensing layer, cloud platform layer, aggregated layer, processing layer

b. Sensing layer, aggregated layer, processing layer, cloud platform layer

c. Aggregated layer, sensing layer, processing layer, cloud platform layer

d. Sensing layer, processing layer, aggregated layer, cloud platform layer

Correct Answer: b. Sensing layer, aggregated layer, processing layer, cloud platform layer

Detailed Solution: The sensing layer senses data and transmit it o the aggregation layer where the data are aggregated. The aggregated layer further transfers the data to the processing layer the data are processed and final sent to the cloud platform. (Please refer Lecture 58@8:25)

QUESTION 8:

Which of the following networks is used in AmbuSens?

a. IEEE 802.15.4

b. 6LoWPAN

c. Wireless Body Area Network

d. IEEE 802.15.2

Correct Answer: c. Wireless Body Area Network

Detailed Solution: In AmbuSens, the network used is Wireless Body Area Network. (Please refer Lecture 58@23:03)

QUESTION 9:

Wireless IoT driven solutions for remote healthcare facility provisioning brings healthcare to patients than bringing patients to healthcare.

a. True

b. False

Correct Answer: a. True

Detailed Solution: In IoT health care, wireless IoT driven solutions bring healthcare to patients than bringing patients to healthcare. (Please refer Lecture 58@11:40)

QUESTION 10:

Fill in the blank. The statistical data analysis tools can be applied on_____.

a. Sensor data

b. Images

c. Videos

d. All of these

Correct Answer: a. Sensor data

Detailed Solution: The statistical data analysis tools can be applied on sensor data. Please refer Lecture 59@18:16)

QUESTION 11:

Which of the following sensors are present in smartphones handheld devices are used for activity monitoring?

a. EEG and GPS

b. Accelerometer and EEG

c. Accelerometer and Gyroscope

d. All of these

Correct Answer: c. Accelerometer and Gyroscope

Detailed Solution: Accelerometer and Gyroscope are used for activity monitoring. (Please refer Lecture 59@15:29)

QUESTION 12:

Which of the following is the primary disadvantage of using camera-based activity tracking?

a. Accurate

b. Processing intensive

c. Both Accurate and Processing intensive

d. None of these

Correct Answer: b. Processing intensive

Detailed Solution: Camera based activity tracking is very much process intensive. (Please refer Lecture 59@17:43)

QUESTION 13:

By performing continuous monitoring of a person's activity, it is not possible to observe his/her behavior or to identify any repetitive pattern in his/her day-to-day activity.

a. True

b. False

Correct Answer: b. False

Detailed Solution: Continuous monitoring of activity results in daily observations of human behavior and repetitive patterns in their activity. (Please refer Lecture 59@14:33)

QUESTION 14:

Which of the following are the inbuilt sensors that are present in high end smartphones?

a. ECG and EEG

b. Accelerometer, proximity sensor, and EEG

c. Accelerometer, Compass, and Gyroscope

d. Pressure sensor and NPK sensor

Correct Answer: c. Accelerometer, Compass, and Gyroscope

Detailed Solution: High end smartphones are likely to have accelerometer, compass, and gyroscope. (Please refer Lecture 60@2:28)

QUESTION 15:

Fill in the blank. Processing the handheld activity device data with artificial intelligence can be used for _____.

a. Fall detection

b. Heart rate detection

c. Vehicle detection

d. All of these

Correct Answer: a. Fall detection

Detailed Solution: Processing the handheld activity device data with artificial intelligence can be used for detecting sudden fall of a person. (Please refer Lecture 60@11:56)

**************END*************